

LewyGym: A Parkinson's Disease Protein Variant Effect Benchmark

Reveals Signal Inversion in Conservation-Based Protein Language Models

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GitHub: github.com/Tyronita/LewyGym * HuggingFace: [EvanOLeary/LewyGym](https://huggingface.co/EvanOLeary/LewyGym) * License: CC0

Protein language models (PLMs) trained on evolutionary sequence conservation achieve strong variant effect prediction on loss-of-function (LOF) proteins, with ESM-2 650M reaching a mean Spearman rho of +0.414 across ProteinGym's 217 benchmark assays. However, ProteinGym contains no Parkinson's disease (PD) proteins, leaving a gap in benchmark coverage for a disease with a well-characterised genetic architecture spanning both LOF and gain-of-function (GOF) mechanisms. We introduce LewyGym, a ProteinGym-compatible benchmark covering the PD gene panel (SNCA, LRRK2, GBA, PRKN, PINK1, PARK7, VPS35). LewyGym provides two evaluation tracks: (i) a DMS fitness track with 13,560 single-residue variants across five alpha-synuclein (SNCA) assays from MaveDB, and (ii) a ClinVar pathogenicity classification track with 421 expert-annotated missense variants across seven PD genes. Baseline evaluation of ESM-2 (8M and 650M parameters) using masked-marginal scoring yields mean Spearman rho of -0.14 and -0.17 on the DMS track -- a marked inversion relative to ProteinGym (+0.414). Larger ESM-2 models amplify rather than correct this inversion, consistent with stronger conservation encoding of a protein where pathogenic variants enhance toxicity rather than disrupt function. LewyGym is released under CC0 with a self-contained scoring script compatible with any HuggingFace masked protein language model.

Keywords: protein language model, Parkinson's disease, variant effect prediction, benchmark, alpha-synuclein, deep mutational scanning, gain-of-function

1. Introduction

Protein language models (PLMs) trained on large protein sequence corpora have demonstrated remarkable zero-shot ability to predict the functional consequences of amino acid substitutions [1]. The dominant evaluation paradigm -- exemplified by ProteinGym [1] and FLIP [2] -- measures Spearman rank correlation between model-predicted log-likelihood ratios and experimentally measured fitness scores from deep mutational scanning (DMS) assays. ESM-2 650M achieves a mean Spearman rho of +0.414 across ProteinGym's 217 substitution assays [1], and ESM-1v reaches +0.440 [3].

This strong performance rests on a key biological assumption: amino acids disfavoured by evolution (low PLM log-likelihood) are also functionally damaging. For loss-of-function (LOF) proteins -- where variants disrupt enzymatic activity, structural stability, or binding affinity -- this assumption holds. The evolutionary record penalises variants that break the protein.

Parkinson's disease (PD) presents a systematic challenge to this assumption. The PD gene panel spans two opposing pathogenic mechanisms. SNCA (alpha-synuclein) and LRRK2 operate through gain-of-function

(GOF): variants that increase alpha-synuclein's aggregation propensity, or hyperactivate LRRK2 kinase, are pathogenic because they enhance a toxic function. For GOF proteins, a residue that promotes aggregation may be evolutionarily disfavoured -- meaning conservation-based PLMs assign it a negative log-likelihood delta, predicting it as evolutionarily unusual, while the DMS fitness score registers harm via a completely different biological mechanism (aggregation-driven toxicity). For LOF genes in the PD panel (GBA, PRKN, PINK1, PARK7), the standard conservation-fitness alignment should hold.

Despite PD affecting more than 10 million people worldwide, ProteinGym contains no PD proteins. No existing benchmark evaluates PLMs across a disease-coherent gene panel spanning both GOF and LOF mechanisms simultaneously. We introduce LewyGym -- named after Lewy bodies, the misfolded alpha-synuclein aggregates that define PD pathology -- to fill this gap. Our baseline results confirm that ESM-2 conservation signals invert on SNCA (mean Spearman rho = -0.14 to -0.17), and that larger ESM-2 models amplify rather than correct this inversion, providing the first systematic evidence that standard PLM benchmarking underestimates model failure on GOF disease proteins.

2. Benchmark Design

2.1 DMS Fitness Track

The DMS track comprises five SNCA assays from MaveDB [7], all measuring yeast fitness as a proxy for alpha-synuclein toxicity. Newberry et al. [5] measured fitness of all single-residue substitutions in a membrane-targeted GFP fusion construct expressed in yeast ($n = 2,728$ variants). Noh et al. [6] measured yeast fitness at four galactose induction concentrations (1%, 0.1%, 0.01%, 0.001%), capturing the concentration-dependent folding and toxicity landscape of alpha-synuclein ($n \sim 2,700$ per assay). All five assays cover the full SNCA sequence (UniProt P37840, $L = 140$). Data were retrieved from MaveDB via the public REST API (api.mavedb.org), converted from HGVS notation (e.g. p.Met1Trp -> M1W) to ProteinGym one-letter substitution format, and normalised to zero-mean unit-variance. Stop codons, insertions, and reference-mismatch variants were excluded (~3-5% per assay).

Assay ID	n	Phenotype	Source (MaveDB)
SNCA_HUMAN_Newberry_2020	2,728	GFP membrane toxicity (yeast)	00000045-k-1
SNCA_HUMAN_Noh_2026_1pct	2,725	Fitness, 1% galactose induction	00001249-a-1
SNCA_HUMAN_Noh_2026_01pct	2,733	Fitness, 0.1% galactose	00001249-a-2
SNCA_HUMAN_Noh_2026_001pct	2,731	Fitness, 0.01% galactose	00001249-a-3
SNCA_HUMAN_Noh_2026_0001pct	2,643	Fitness, 0.001% galactose	00001249-a-4
Total	13,560		

2.2 ClinVar Pathogenicity Track

The pathogenicity track provides ClinVar-classified missense variants for all seven PD genes: SNCA, LRRK2, GBA, PRKN, PINK1, PARK7, and VPS35. Only missense single-nucleotide variants with at least one submitter and no conflicting interpretations are retained. After deduplication on (mutant, label) pairs, 421 variants remain: 314 Pathogenic/Likely Pathogenic (P/LP) and 107 Benign/Likely Benign (B/LB). Table 2 shows per-gene counts. GBA dominates ($n = 240$) due to the large literature on GBA variants in Gaucher disease, the strongest known genetic risk factor for PD. The evaluation metric is AUROC; model scores are negated before ranking (more negative log-likelihood delta = more damaging).

Gene	Pathogenic mechanism	P/LP	B/LB	n
GBA	LOF -- glucocerebrosidase (GD/PD risk)	237	3	240
LRRK2	GOF -- kinase hyperactivation	10	71	81
PRKN	LOF -- ubiquitin E3 ligase	28	14	42
PINK1	LOF -- mitophagy kinase	21	11	32
PARK7	LOF -- oxidative stress sensor	10	3	13
SNCA	GOF -- aggregation (hallmark PD)	5	3	8
VPS35	GOF -- retromer complex dysfunction	3	2	5
Total		314	107	421

2.3 Format and Compatibility

All DMS files follow ProteinGym substitution CSV format (columns: mutant, DMS_score). Wildtype sequences, gene metadata, and assay phenotype descriptions are provided in reference_files/DMS_substitutions.csv. The scoring script score.py implements masked-marginal scoring (Section 3.1) and accepts any HuggingFace EsmForMaskedLM checkpoint. Dependencies: transformers, torch, pandas, scipy, numpy. The full DMS track can be reproduced from scratch in under 2 minutes on a CPU for ESM-2 8M, and under 30 minutes for ESM-2 650M.

3. Baseline Evaluation

3.1 Masked-Marginal Scoring

We use masked-marginal scoring [3], the standard ProteinGym evaluation protocol. For each position i in the wildtype sequence (1-indexed, excluding special tokens), position i is masked and a single forward pass computes the log-softmax distribution over the amino acid vocabulary. The score for variant $X \rightarrow Y$ at position i is:

$$\text{score}(X \rightarrow Y) = \log p(Y \mid \text{seq with position } i \text{ masked}) - \log p(X \mid \text{seq with position } i \text{ masked})$$

For multi-mutant variants, per-position log-likelihood deltas are summed. Building the position cache requires L forward passes ($L=140$ for SNCA), approximately 1-2 seconds on CPU for ESM-2 8M and 20-30 seconds for ESM-2 650M. Spearman rank correlation between model scores and experimental DMS fitness values is the primary DMS track metric.

3.2 Models Evaluated

We evaluate two ESM-2 checkpoints [4]: 8M parameters (facebook/esm2_t6_8M_UR50D, 6 transformer layers, 320-dim embeddings) and 650M parameters (facebook/esm2_t33_650M_UR50D, 33 layers, 1280-dim). ESM-2 650M is the largest checkpoint that achieves peak ProteinGym performance ($\rho = +0.414$) before diminishing returns at 3B and 15B parameters [1]. Both models use 32-bit float precision and SDPA attention.

3.3 DMS Track Results

Assay	ESM-2 8M	ESM-2 650M	n
SNCA_HUMAN_Newberry_2020	-0.165	-0.210	2,728
SNCA_HUMAN_Noh_2026_1pct	-0.147	-0.168	2,725

SNCA_HUMAN_Noh_2026_01pct	-0.135	-0.160	2,733
SNCA_HUMAN_Noh_2026_001pct	-0.115	-0.146	2,731
SNCA_HUMAN_Noh_2026_0001pct	-0.134	-0.171	2,643
Mean	-0.139	-0.171	13,560

Table 3. Negative Spearman rho values indicate signal inversion: model conservation scores anti-correlate with experimental fitness. For comparison, ESM-2 650M achieves +0.414 mean Spearman rho on ProteinGym’s 217 LOF assays [1]. The 650M model amplifies the inversion vs 8M (-0.171 vs -0.139) in four of five assays.

All five SNCA assays yield negative Spearman correlations for both models. ESM-2 650M produces a more negative mean correlation than 8M in four of five assays (Table 3), with the largest inversion on the Newberry 2020 membrane toxicity assay ($\rho = -0.210$). The mean DMS track score is -0.139 (8M) and -0.171 (650M), compared to the ProteinGym LOF baseline of +0.414 for ESM-2 650M [1] -- a departure of -0.585 Spearman units, exceeding the full per-assay variation range reported for ESM-2 on ProteinGym.

4. Discussion

The negative Spearman correlations on all five SNCA DMS assays arise from a mechanistic mismatch between conservation-based scoring and GOF pathogenicity. ESM-2 is trained to reconstruct masked residues from evolutionary context: residues conserved across UniRef50 receive high log-likelihood, and unusual substitutions receive low log-likelihood. For SNCA, the positions most critical for aggregation-driven toxicity are evolutionarily constrained. The model correctly identifies them as evolutionarily disfavoured, producing negative log-likelihood deltas for aggregation-promoting variants -- yet in the yeast DMS assay, these same variants reduce fitness (increase toxicity), and the correlation inverts. In contrast, substitutions that disrupt aggregation-prone contacts are evolutionarily unusual but improve yeast fitness, further inverting the signal.

The amplification of the inversion by larger ESM-2 models (650M > 8M) is mechanistically consistent: larger models learn richer conservation representations and express higher confidence in penalising evolutionarily unusual residues. As SNCA’s GOF-relevant variants fall precisely in the ‘unusual but toxic via aggregation’ category, a more confident conservation model produces a stronger inversion signal. This is distinct from the behaviour on ProteinGym, where larger ESM-2 models improve LOF prediction because deeper conservation encoding better captures functional constraint.

These results have three practical implications. First, conservation-based PLMs should not be used as zero-shot GOF pathogenicity predictors without recalibration or task-specific fine-tuning. Second, aggregate benchmarks like ProteinGym -- which necessarily average across heterogeneous proteins -- may obscure systematic failures on GOF disease proteins. Third, disease-specific benchmarks spanning both GOF and LOF genes within a coherent disease panel are essential for characterising model behaviour on clinically actionable variants.

Limitations: LewyGym v0.1 includes DMS data for SNCA only. LRRK2, GBA, and other PD genes lack large-scale publicly available DMS assays. The ClinVar pathogenicity track has class imbalance (GBA, $n=240$, dominates) and very small variant counts for some GOF genes (SNCA $n=8$, VPS35 $n=5$). Planned additions for v0.2 include LRRK2 kinase activity assays (~70 variants curated from literature), GBA enzyme activity data (~50 variants), and baselines for ESM-1v [3], AlphaMissense [8], and Evo2 [9].

5. Data Availability

All data, code, and results are publicly available at github.com/Tyronita/LewyGym and huggingface.co/datasets/EvanO'Leary/LewyGym under CC0 (public domain). DMS data are redistributed from MaveDB under their original CC0 licences. ClinVar data are in the public domain. The scoring script `score.py` reproduces all baseline results in Table 3 from the raw CSV files.

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References

- [1] Notin, P. et al. ProteinGym: Large-Scale Benchmarks for Protein Fitness Prediction and Design. NeurIPS Datasets and Benchmarks 2023. doi:10.1101/2023.12.07.570727
- [2] Dallago, C. et al. FLIP: Benchmark tasks in fitness landscape inference for proteins. NeurIPS Datasets and Benchmarks 2022.
- [3] Meier, J. et al. Language models enable zero-shot prediction of the effects of mutations on protein function. NeurIPS 2021. doi:10.1101/2021.07.09.450648
- [4] Lin, Z. et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. Science 379, 1123-1130 (2023). doi:10.1126/science.ade2574
- [5] Newberry, R.W. et al. Deep mutational scanning reveals the structural basis of alpha-synuclein activity. Nature Chemical Biology 16, 653-659 (2020). doi:10.1038/s41589-020-0480-6
- [6] Noh, J. et al. Concentration-dependent fitness landscape of alpha-synuclein. Protein Science 2026. doi:10.1002/pro.70456
- [7] Esposito, D. et al. MaveDB: an open-source platform to distribute and interpret data from multiplexed assays of variant effect. Genome Biology 20, 223 (2019). doi:10.1186/s13059-019-1845-6
- [8] Cheng, J. et al. Accurate proteome-wide missense variant effect prediction with AlphaMissense. Science 381, eadg7492 (2023). doi:10.1126/science.adg7492
- [9] Nguyen, E. et al. Sequence modeling and design from molecular to genome scale with Evo 2. bioRxiv 2024. doi:10.1101/2024.02.27.582234